

Layered Sector Rotation Methodology

A multi-timeframe, evidence-based framework for identifying bullish sectors and alerting on bearish reversals across US sectors, international markets, and style factors.

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1. Executive summary

There is no single "proven" methodology — there are seven pillars that *each* have peer-reviewed or 30+ years of out-of-sample evidence behind them. Used together, they are highly redundant and that redundancy is the source of their robustness.

The pillars are:

#	Pillar	Source	What it tells you
1	Cross-sectional momentum (12-1)	Jegadeesh & Titman 1993; Moskowitz & Grinblatt 1999	Which sectors have been outperforming peers over the past 12 months (excluding the last month to skip short-term reversal)
2	Time-series momentum / Faber 10-month SMA	Faber 2007; Moskowitz, Ooi & Pedersen 2012	Whether each sector is in its own uptrend (binary risk-on/off)
3	Weinstein Stage Analysis	Weinstein 1988	Whether the sector is in Stage 2 (advance) vs Stage 1/3/4 — uses 30-week MA + Mansfield RS
4	Antonacci Dual Momentum	Antonacci 2014	Combines absolute (vs T-bill) + relative (vs peers) momentum — the absolute filter is the catastrophic-loss circuit-breaker
5	Relative Rotation Graphs (RRG)	de Kempenaer 2004/2011	Where each sector is in the rotation cycle: Leading → Weakening → Lagging → Improving

3. Universe

3.1 US sectors (11 SPDRs)

XLK XLF XLE XLV XLI XLY XLP XLU XLB XLRE XLC

3.2 US industries (drill-down, optional Phase 2)

SOXX IGV IHI IHF KRE KIE ITB XHB XRT XOP OIH KWEB FDN ITA JETS SMH IBB XBI GDX TAN ICLN

3.3 International / country & region

VEU VWO EFA EEM EWJ EWG EWU INDA MCHI FXI EWZ EWA EWC EWW KSA EZA EIDO EWY EWT EWS

3.4 Style factors (US)

MTUM QUAL USMV VLUU SIZE IWF IWD IJR IJH IWB

3.5 Defensive / absolute-momentum bench

BIL (T-bill proxy, for Antonacci absolute filter) · TLT IEF GLD UUP DBC (risk-off candidates)

3.6 Benchmark

SPY for US, ACWI for global, URG for developed-world cross-sector comparison

4. Indicator formulas

All inputs are weekly bars (Friday close) unless noted; daily bars only for Layer 3 triggers. Use **adjusted close** (total return) so dividends flow through.

4.1 Cross-sectional momentum (Jegadeesh-Titman 12-1)

$$M1_i = (P_i[t-21] / P_i[t-252]) - 1$$

where t is today (daily bars) and 252 trading days $\approx$ 12 months, skip 21 days (1 month) to remove short-term reversal. Compute for each ticker i in the universe. Rank cross-sectionally; the top decile is the "winner" basket.

4.2 Time-series momentum (Faber 10-month SMA)

$$\begin{aligned} \text{SMA10}_i &= \text{mean}(\text{monthly_close}_i[t-9 \dots t]) \\ \text{TS_signal}_i &= 1 \text{ if } \text{monthly_close}_i[t] > \text{SMA10}_i \text{ else } 0 \end{aligned}$$

Apply on monthly bars. This is the binary "is this asset itself in an uptrend" filter.

4.3 Weinstein Stage 2 (weekly)

Inputs on weekly bars:

$$\text{SMA30w} = \text{mean}(\text{weekly_close}[t-29 \dots t])$$

```
slope = ( SMA30w[t] - SMA30w[t-5] ) / 5      # positive slope over last 5 weeks
mansfield_RS = ( (P_i[t] / P_bench[t]) / (P_i[t-52]/P_bench[t-52]) - 1 ) * 100
```

Stage 2 (bullish) is true when:

```
P_i[t] > SMA30w AND slope > 0 AND mansfield_RS > 0
```

Stage 4 (bearish) is true when:

```
P_i[t] < SMA30w AND slope < 0 AND mansfield_RS < 0
```

4.4 Antonacci dual-momentum filter

```
abs_mom_i = total_return_i[252 trading days] - total_return_BIL[252 trading days]
A_pass_i = 1 if abs_mom_i > 0 else 0
```

A sector cannot be a buy unless A_pass_i = 1. This is the catastrophic-loss circuit-breaker.

4.5 RRG — RS-Ratio and RS-Momentum

de Kempnaer's exact formulas are proprietary, but a widely-published approximation is:

```
RS_raw_i      = 100 * P_i / P_bench
RS_ratio_i    = 100 + ( RS_raw_i - SMA(RS_raw_i, 63) ) / stdev(RS_raw_i, 252) * 10
RS_momentum_i = 100 + ( RS_ratio_i - SMA(RS_ratio_i, 5) ) / stdev(RS_ratio_i, 21) * 10
```

Plot RS-Ratio on X, RS-Momentum on Y, axes cross at 100. Quadrants:

```
Leading    = RS_ratio > 100 AND RS_momentum > 100
Weakening = RS_ratio > 100 AND RS_momentum < 100
Lagging    = RS_ratio < 100 AND RS_momentum < 100
Improving = RS_ratio < 100 AND RS_momentum > 100
```

4.6 Sector breadth (daily)

For each sector ETF, compute the percentage of its top-N holdings (or sector-index constituents) above their 50-day SMA:

```
B50_sector = mean( 1 if P_j > SMA50_j else 0 for j in constituents )
```

- Bullish: B50 >= 60%
- Neutral: 40% <= B50 < 60%
- Bearish: B50 < 40%

If constituent-level data is unavailable, substitute the McClellan-style oscillator on the sector ETF itself.

4.7 Institutional money flow & volume — the "smart money" pillar

This is the layer that tells you whether real institutional capital is moving in or out of a sector — not just whether price is moving. Price can rise on light volume (suspect) or on heavy institutional sponsorship (confirmed). The metrics below are the canonical ones with peer-reviewed or industry-standard backing.

4.7.1 On-Balance Volume (OBV) — Granville 1963

The earliest accumulation/distribution indicator. Adds the day's volume to a running total on up days, subtracts on down days.

```
OBV[t] = OBV[t-1] + sign(close[t] - close[t-1]) * volume[t]
OBV_slope_20 = linreg_slope( OBV[t-19 .. t] ) # 20-day regression slope
```

Bullish: OBV is making new highs *with* or *ahead* of price.

Bearish (divergence): Price makes a new high but OBV does not.

4.7.2 Chaikin Money Flow (CMF, 21-period) — Chaikin 1981

The single best volume-weighted accumulation/distribution measure for an ETF. Combines where each bar closes within its range with the volume on that bar.

```
MFM = ((close - low) - (high - close)) / (high - low) # money-flow multiplier, range [-1, +1]
MFV = MFM * volume # money-flow volume
CMF21 = sum( MFV[t-20 .. t] ) / sum( volume[t-20 .. t] )
```

Reading:

- CMF21 > +0.10 → strong accumulation (institutional buying pressure)
- +0.05 to +0.10 → mild accumulation
- -0.05 to +0.05 → neutral
- -0.05 to -0.10 → mild distribution
- CMF21 < -0.10 → strong distribution (institutional selling)

4.7.3 Money Flow Index (MFI, 14-period) — "volume-weighted RSI"

```
typical_price = (high + low + close) / 3
raw_money_flow = typical_price * volume
positive_mf = sum of raw_money_flow on days where typical_price > yesterday's
negative_mf = sum of raw_money_flow on days where typical_price < yesterday's
money_ratio = positive_mf / negative_mf # over the 14-day window
MFI = 100 - (100 / (1 + money_ratio))
```

- MFI > 80 overbought (often near distribution tops)
- MFI < 20 oversold (often near accumulation bottoms)
- **MFI/price divergence** is one of the cleanest reversal signals there is.

4.7.4 Relative Volume (RVOL) — unusual volume detector

The "tape is telling you something" indicator.

```
RVOL[t] = volume[t] / mean( volume[t-20 .. t-1] )
```

- RVOL ≥ 1.5 → unusual (something happened)
- RVOL ≥ 2.0 → highly unusual (news, earnings, institutional sweep)
- RVOL ≥ 3.0 → block-print suspect; investigate the tape

4.7.5 Block trades & dark-pool flow — institutional footprints

A "block" is a single print of ≥ 10,000 shares or ≥ \$200,000 notional (NYSE Rule 127 / FINRA Rule 6240). Most institutional executions are split across many lit prints and printed off-exchange ("dark"); the dark-pool share is reported via FINRA ATS data.

```
block_count[t] = count of single prints ≥ 10,000 shares
```

```

block_notional[t] = sum of notional value of those prints
dark_pool_pct[t] = off_exchange_volume[t] / consolidated_volume[t] # 0..1
upside_block_ratio = block_notional_on_uptick / block_notional_on_downtick # > 1 = bullish

```

Bullish reading:

- $\text{dark_pool_pct} \geq 45\%$ *with* $\text{upside_block_ratio} > 1.5$ for ≥ 3 sessions → silent institutional accumulation
- Sustained block_count above 20-day average

Bearish reading:

- $\text{upside_block_ratio} < 0.7$ while price is flat or rising → distribution into strength
- A surge in block trades on a *down* day at the end of a Stage 2 advance is a classic Stage 3 distribution signal

Data sources: FINRA OTC Transparency (free, T+1), Cboe BZX/EDGX, NYSE TAQ (paid), or aggregators like FlowAlgo, Quiver Quant, Cheddar Flow.

4.7.6 Trade velocity / pace — speed of institutional execution

"Velocity" measures how fast institutions are accumulating relative to the recent baseline.

```

trade_count[t] = number of executions
notional_velocity = ( dollar_volume[t] / trade_count[t] ) / sma( dollar_volume / trade_count, 20 )

```

A rising **average trade size** (notional / count) combined with rising volume signals that the average market participant is getting larger — i.e., funds, not retail. Conversely, falling trade size on rising volume often signals retail capitulation or distribution.

4.7.7 ETF primary-market flows — creations & redemptions

For sector ETFs (XLK, SOXX, etc.) this is the cleanest possible institutional-flow signal because it requires an authorized participant to deliver baskets of underlying shares.

```

shares_outstanding_change[t] = SH0[t] - SH0[t-1] # from issuer daily file
estimated_net_flow[t] = shares_outstanding_change[t] * NAV[t]
nf_5d_pct = sum( estimated_net_flow[t-4 .. t] ) / AUM[t] # net flow last 5d as % of AUM

```

- $\text{nf_5d_pct} > +1.5\%$ → strong primary inflow (real money entering the sector basket)
- $\text{nf_5d_pct} < -1.5\%$ → strong primary outflow (money exiting)

Issuer daily holdings/SHO files are free: iShares, SSGA, Invesco, Vanguard publish them.

4.7.8 13F institutional positioning (quarterly, lagged)

SEC 13F filings (45-day lag) reveal which sectors hedge funds and institutions added to. Aggregate across the largest 100 active managers:

```

hf_net_buys_sector_q = Σ ( current_q_shares - prior_q_shares ) × avg_price for each sector

```

- Sectors with positive Q-over-Q net buys from a majority of top managers ($\geq 60\%$ of the 100) have a structural tailwind for the next 1–2 quarters.

- The data is lagged but the *direction* persists; useful as a slow-moving confirmation of the cross-sectional momentum signal.

Sources: SEC EDGAR (free), WhaleWisdom, 13F.info, Bloomberg.

4.7.9 Short interest & days-to-cover

Bi-monthly FINRA Reg SHO data. Rising short interest in a Stage-2 advancing sector is fuel for a short-squeeze; falling short interest while price rises is "real" buying.

```
SI_ratio[t]      = short_interest[t] / shares_outstanding[t]
days_to_cover  = short_interest[t] / 20d_avg_daily_volume
SI_delta_15d    = SI_ratio[t] - SI_ratio[t-15]  # change over the half-month
```

- $SI_delta_15d < 0$ while price rising $\rightarrow$ genuine demand absorbing supply
- $SI_delta_15d > 0$ while price falling $\rightarrow$ distribution + bearish positioning
- $days_to_cover > 5$ in a rising sector $\rightarrow$ squeeze potential

4.7.10 Composite "institutional flow" score per sector

Aggregate the above into a single z-scored composite that can be used in the master ranking:

```
F_i = 0.30 * z(CMF21_i)           # primary accumulation/distribution
      + 0.20 * z(OBV_slope_20_i)  # volume-trend confirmation
      + 0.20 * z(nf_5d_pct_i)      # ETF primary flow (or 13F if no ETF flow)
      + 0.10 * z(upside_block_ratio_i) # block-trade direction
      + 0.10 * z(notional_velocity_i) # trade-size velocity
      + 0.10 * (-1) * z(SI_delta_15d_i) # short-interest improvement (sign-flipped)
```

F_i is the **Pillar 7 institutional flow score** and enters the master composite in §5.

4.7.11 Hard veto rule

Even with a perfect price-based score, a sector cannot be added if $F_i < -0.5\sigma$ (i.e., it's in the bottom ~30% of the universe on flow). This prevents buying "price-only" rallies that aren't backed by real money — a signature failure mode of pure-momentum systems.

4.8 Business-cycle phase (monthly)

Use a deterministic mapping:

```
phase = EARLY    if ISM_PMI > 50 and rising and yield_curve > 0
          MID     if ISM_PMI > 50 and slowing and yield_curve > 0
          LATE    if ISM_PMI > 50 and yield_curve < 0
          RECESSION if ISM_PMI < 50 (2 consecutive months)
```

ISM PMI from FRED (series MANEMP or use Markit PMI as fallback), yield curve = 10Y - 2Y from FRED (T10Y2Y).

Map phase $\rightarrow$ favored US sectors (Fidelity Business Cycle table):

- **Early:** XLY, XLF, XLRE, XLI, XLK, XLB
- **Mid:** XLK, XLC, XLI (sector dispersion is lowest in this phase)
- **Late:** XLE, XLB, XLP, XLV
- **Recession:** XLP, XLU, XLV

The macro overlay does *not* override sector signals — it adjusts the composite weight (Section 5) by +10% for in-phase sectors and -10% for out-of-phase sectors.

5. Composite scoring (Layer 2 output)

Per ticker, per week (Friday close):

```

z(x)          = cross-sectional z-score of x across the active universe
filters_i     = TS_signal_i + Stage2_i + A_pass_i    # 0..3

S_i = 0.22 * z(M1_i)                # 12-1 momentum
      + 0.12 * z(mansfield_RS_i)     # Weinstein RS
      + 0.15 * z(RS_ratio_i)         # RRG horizontal
      + 0.08 * z(RS_momentum_i)      # RRG vertical
      + 0.12 * (filters_i / 3)        # binary trend filters (TS, Stage2, Antonacci)
      + 0.08 * cycle_tilt_i           # +1 / 0 / -1 from business-cycle map
      + 0.23 * z(F_i)                # *** institutional flow composite ($4.7.10) ***

# Hard veto: drop any sector where F_i < -0.5σ even if S_i is high
# (price-only rallies without flow are rejected)

# Hold top N
N_us_sectors   = 4    (out of 11)
N_industries   = 3    (out of ~20)
N_countries    = 3    (out of ~20)
N_factors      = 1-2  (out of 10)

```

Why flow gets ~23% of the weight — it's an independent confirmation of the price-based pillars. When price and flow agree, signal quality is much higher; when they disagree, the trade is almost always wrong. Empirically (Lee & Swaminathan 2000; Chordia & Swaminathan 2000) volume-confirmed momentum substantially outperforms pure-price momentum on a risk-adjusted basis.

Rebalance weekly (Friday close → execute Monday open). The 12-1 momentum literature is unanimous that monthly rebalancing is fine; weekly is the *fastest* rebalance that still respects the academic evidence. Daily rebalancing destroys the edge in transaction costs.

6. Bearish alert state machine (the part you specifically asked about)

Each held position has a state. The state advances on the worst signal across timeframes; it can only *retreat* (improve) after 4 consecutive weekly closes of improvement.

State	Entry condition (any one trips advance)	Action	Alert priority
STAGE_2_BULLISH	All Stage-2 conditions + RRG Leading + B50 ≥ 60% + CMF21 > +0.05 + nf_5d_pct > 0	Buy / Add	n/a

State	Entry condition (any one trips advance)	Action	Alert priority
HOLD	Stage-2 intact, CMF21 still > 0, no flow divergence	Hold	none
WARNING	RRG drops to Weakening ≥ 2 wks, OR breadth B50 < 50%, OR M1 rank out of top-tercile, OR CMF21 < 0 for 2 wks , OR price/OBV bearish divergence , OR upside_block_ratio < 1.0 for 3+ sessions	Tighten stop to 30wMA, no new adds	LOW (info)
EXIT	Weekly close below 30wMA, OR Mansfield RS turns negative, OR Antonacci A_pass = 0, OR RRG enters Lagging, OR CMF21 < -0.10 , OR nf_5d_pct < -1.5% (sustained ETF redemptions) , OR block tape flips: upside_block_ratio < 0.7 for 5 sessions	Sell on Monday open, send alert	HIGH (action)
BEARISH / STAGE_4	EXIT condition + 30wMA slope negative + RRG Lagging ≥ 3 wks + CMF21 < -0.10 confirmed for 3 wks	Avoid; short candidate	HIGH (action)
STAGE_1_BASING	Came from BEARISH; price reclaimed 30wMA but slope still flat AND CMF21 turns positive	Watch for re-entry	LOW (info)

Flow-only early-warning alerts (fire even before state advances):

- **DISTRIBUTION_DAY**: large up-volume day ($RVOL \geq 1.5$) closing in the lower 25% of its range — classic O'Neil distribution signal. 4+ in 25 sessions historically precedes Stage-3 tops.
- **DARK_POOL_SELL**: dark_pool_pct > 50% with price flat or down for 3+ days — silent institutional exit.
- **OBV_NEGATIVE_DIVERGENCE**: price makes a new 20-day high but OBV does not — early Stage 3 warning.

6.1 Alert delivery

Three escalation levels:

- **LOW** — daily digest email at 8am ET (states changed yesterday)
- **HIGH** — immediate push (Pushover / Telegram / Slack webhook) on EXIT or BEARISH entry
- **MACRO** — separate channel when Faber 10-month SMA flips on the SPY benchmark (whole-portfolio risk-off)

6.2 Cooldown

- A position that exited cannot be re-bought for 4 weeks unless it goes BEARISH → STAGE_1 → STAGE_2 sequence. This prevents whipsaws.

7. Data, libraries, scheduling

7.1 Data sources

Need	Free option	Paid option
Daily/weekly OHLC ETF	yfinance (Yahoo)	Polygon, Tiingo, EOD Historical Data
Macro (PMI, yield curve, CPI)	fredapi (FRED) — free	Bloomberg, Refinitiv
Sector constituents	iShares / SSGA holdings CSVs (free)	FactSet, S&P Capital IQ
ETF SHO / primary flow	Issuer daily holdings files (iShares, SSGA) — free	Bloomberg, ICI
Short interest	FINRA Reg SHO bi-monthly (free)	S3 Partners, Markit
13F institutional holdings	SEC EDGAR (free, T+45)	WhaleWisdom, 13F.info, Bloomberg
Block trades / lit prints	NYSE TAQ delayed (free), Polygon trade-tape	Refinitiv MarketPsych, NYSE TAQ realtime
Dark pool % / ATS volume	FINRA ATS Transparency (free, T+1)	FlowAlgo, Cheddar Flow, Quiver Quant
Unusual options activity (proxy for flow)	CBOE delayed feeds	Unusual Whales, BlackBoxStocks
Backtest engine	vectorbt, zipline-reloaded	bt, Backtrader is free too

7.2 Python stack

```
python ≥ 3.11
pandas, numpy, scipy
yfinance, fredapi, pandas-datareader
```

pandas-ta or ta-lib for indicators (CMF, MFI, OBV all included)
 sec-edgar-downloader # 13F filings
 finra-data # short interest, ATS dark pool volumes
 requests + lxml # iShares/SSGA SHO daily CSV ingestion
 matplotlib + plotly for RRG visualization
 vectorbt (or bt) for backtests
 apscheduler (or cron + systemd) for scheduling
 requests for webhook alerts
 sqlite (or DuckDB) for the state-machine persistence

7.3 Job schedule (cron, US Eastern)

Job	Frequency	Time
update_prices.py	daily	18:00 ET (after close + reconciliation)
compute_daily_triggers.py	daily	18:15 ET
compute_weekly_signals.py	weekly	Fri 18:30 ET
compute_monthly_macro.py	monthly	1st business day, 09:00 ET
state_machine_advance.py	daily	18:20 ET
send_alerts.py	daily + on-trigger	07:55 ET digest; immediate on HIGH

7.4 Repository layout

```

sector-rotation/
├── data/
│   ├── ohlc_daily.duckdb # price history
│   ├── fred.duckdb      # macro
│   └── state.json        # state machine snapshot
├── src/
│   ├── universe.py       # the tickers in §3
│   ├── indicators.py     # formulas in §4
│   ├── scoring.py        # composite in §5
│   ├── state_machine.py  # transitions in §6
│   ├── alerts.py         # webhook senders
│   └── backtest.py       # vectorbt harness
├── jobs/                 # cron entry points
├── notebooks/            # research / RRG plots
└── tests/
  
```

8. Backtest plan (before going live)

- Data range:** 2003–today (so it spans the 2008 GFC, 2011 EU crisis, 2020 COVID, 2022 bear). Use longest possible for ETFs that exist; substitute index total-return for older history (e.g., MSCI EAFE for EFA).
- Walk-forward:** Train universe weights on 2003–2014, lock parameters, then run 2015–today as out-of-sample. Do **not** tune on the out-of-sample window.
- Benchmark:** 60/40 SPY/AGG and equal-weight 11-sector portfolio.

4. **Metrics to log:** CAGR, Sharpe, Sortino, max drawdown, Calmar, hit-rate by signal, turnover, average holding period, transaction-cost sensitivity (3, 5, 10 bps per side).

5. **Stress checks:**

- Run with the 2-month and 6-month lookbacks instead of 12-1 — does CAGR collapse? (If yes, you've overfit to 12-1.)
- Remove the Antonacci absolute filter — does max drawdown blow out? (Antonacci's own data says yes — it should ~halve drawdown.)
- Shuffle the universe (random sectors): does CAGR fall to benchmark? (Sanity check that the signal is doing work.)

6. **Acceptance criteria** to deploy live:

- Out-of-sample Sharpe ≥ 0.7 (60/40 historical is ~ 0.4)
- Max drawdown $\leq 75\%$ of equal-weight benchmark drawdown
- Turnover $\leq 300\%$ annual (otherwise costs eat returns)
- State machine transitions ≤ 4 per ticker per year on average

9. Pitfalls & known weaknesses

1. **Momentum crashes** (Daniel & Moskowitz 2016): pure 12-1 momentum has rare but brutal reversal months. The Antonacci absolute filter + Weinstein 30wMA exit *mostly* prevents these. Backtest must include 2009-Apr (the canonical momentum-crash month).

2. **"Myth of sector rotation"** (Molchanov & Stangl 2018): the deterministic business-cycle $\rightarrow$ sector map underperforms once you account for forward-looking phase identification. **Conclusion:** the macro overlay should only nudge the composite, not drive it.

3. **RRG is a visualization, not a system:** the RS-Ratio/RS-Momentum formulas above are an approximation; the proprietary version differs. Don't claim RRG-branded backtests if you publish.

4. **Data-mined factor zoo:** limit yourself to the five canonical factors (momentum, value, quality, low-vol, size) — anything else is suspect.

5. **Survivorship bias** in ETF universe: a few sector / country ETFs have closed (e.g., GREK relisted multiple times). Use a delisted-aware data provider for honest backtests.

6. **Bear-market false positives:** the state machine will whipsaw in a choppy market. The 4-week cooldown (\$6.2) and "2-of-3 timeframes" rule (\$2) are designed for this — don't relax them.

10. Build phases (suggested 9-week plan)

Week	Deliverable
1	Data pipeline: yfinance ingest + FRED macro + DuckDB schema

Week	Deliverable
2	Indicators module: §4.1–4.6 + 4.8 with unit tests against known values
3	Institutional-flow module (§4.7): CMF, OBV, MFI, RVOL; iShares SHO ingestion
4	Block-trade & dark-pool feed (FINRA ATS + Polygon tape, or paid feed); 13F + short interest ingestion
5	Scoring module (§5) + ranking output as CSV + RRG notebook
6	State machine (§6) with state persistence + flow-veto + distribution-day alerts
7	Alerting (webhook to your preferred channel) + cron jobs
8	Backtest (§8) — full walk-forward, with/without flow pillar
9	Paper-trade dashboard (Streamlit or simple HTML); go-live decision gate

11. References

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